

- 6 The table below shows the observations when a hot wire is inserted into separate samples of hydrogen chloride, hydrogen bromide and hydrogen iodide gas.

hydrogen halide	observation
HCl	no observable change
HBr	reddish brown vapour only when wire is very hot
HI	purple vapour immediately forms

Which statements explain the observations?

- 1 Valence orbital size increases from Cl to Br to I.
- 2 Bond strength decreases from H-Cl to H-Br to H-I.
- 3 The electron cloud size of the hydrogen halide increases from HCl to HBr to HI.

A 1, 2, and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 1 only